

Adjudication First: A Reproducible Scoreboard for Auto-Renormalization in Weather Modeling

LSWM/ARM: ScaleBench, a Live AIFS Comparison Harness, and Honest Lessons from Real-Data Training Rings

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Draft v0.5 — September 28, 2026

(Numbers are live-measured and roll forward; see §8 for the update mechanism.)

Abstract

Weather AI currently has no reproducible public scoreboard: headline scores are issued by each team on its own data, calibration, and truth conventions, and the community cannot recompute them. We take the opposite position — *adjudication first*. Before claiming a better engine, we built the instrument that makes such claims publicly checkable. (i) **ScaleBench** (v1.4), a public benchmark for *auto-renormalization*: given a microscopic simulator, can a machine discover macroscopic effective laws and honestly report their validity boundaries and error bars? Five task families map onto the five canonical failure modes of learned effective theories (textbook recitation, grid overfitting, short-horizon fitting, interpolation masquerading as law, and chaos-unaudited pseudo-laws); the current deterministic leaderboard scores 0.9864 overall on twenty tasks and is executable-arbitrated end-to-end by `reproduce.py` (external submissions require full reproduction within ± 0.02). (ii) A **live comparison harness** against ECMWF’s AIFS: a rolling sha256-manifested archive and a measured baseline table (12–120 h, z_{500} /MSL RMSE, global and Pearl-River-Delta window), with truth calibers explicitly labeled (self-consistency vs. independent reanalysis) and run-to-run spread recorded as first-class data. (iii) **Real-data training rings** (ERA5-driven hybrid core with through-solver SPSA): the online-training protocol improves on its own objective on two independent truth channels, while the absolute skill gap between them is attributed — by measured rejection of the scale hypothesis — to dynamics off the model manifold rather than to unresolvable wavelengths. (iv) **Physicalization of the rain–flood chain**: ERA5 total precipitation drives the shallow-water organ in physical millimetres with one cross-case calibration, abolishing per-case normalization. Throughout, we apply one rule: every number is measured, every failure is registered (including one of our own retractions), and every physical claim ships with an analytic reference test. The benchmark, the leaderboard JSON, the reproduction script, and the rolling measured tables are public.

1 Introduction: weather AI has no scoreboard

Operational AI weather models now rival or exceed NWP on aggregate skill metrics [1, 2, 3]. Yet the evidence is produced under each team’s private protocol: different verification periods, different truth streams (own analysis vs. ERA5 vs. observations), different regional weighting, and no public mechanism to recompute a published number. The situation mirrors pre-ImageNet computer vision and pre-GLUE NLP — and the remedy that worked there was not a better model but a *better ruler*.

A second, deeper gap sits underneath. The grand promise of learned weather models is implicit *renormalization*: from resolved-scale dynamics, infer the effective equations of the scales below. But “learns sub-grid physics” is an unfalsifiable slogan unless one can *examine* the learned effective theory: does it recover known laws where the answer is known? does it stay stable a hundred training-horizons out? does it extrapolate, or merely interpolate? does it know when it does not know?

This paper is the scoreboard for both gaps, and the first measured season of playing by it. Our contributions:

1. **ScaleBench v2.3** (§3): a public, anti-gaming benchmark for auto-renormalization with an executable arbitration script; leaderboard overall 0.9864 (20 tasks), deterministic pipeline.
2. **A live AIFS comparison harness** (§4): rolling archive with byte-level provenance, measured deterministic baselines at 12–120 h with explicit truth calibers, and a precipitation ETS channel whose first *valid* independent-truth sample landed 2026-09-24 (ETS 0.712 at 0.1 mm threshold); ENS endpoints probed and reported absent from the open store.
3. **Real-data training rings** (§5): ERA5-driven online training works on two truth channels; the skill deficit on the cleaner channel is diagnosed as dynamical, not spectral, via a registered negative result and a balance-manifold projection audit.
4. **Physicalized rain–flood coupling** (§6): ERA5 tp drives the flood organ in physical mm with a single cross-case calibration; per-case normalization abolished and guarded against regression.
5. **Discipline as method** (§7): analytic reference tests for all physics code, registered failures instead of loosened thresholds, and a public retraction of one of our own mismatched measurements.

2 The ARM architecture in one page

The Auto-Renormalization Machine (ARM) is a three-loop engine. The *inner loop* learns closures for a simulator’s sub-grid tendencies; the *middle loop* learns coarse-graining maps (which macroscopic variables exist); the *outer loop* discovers effective laws and emits them as **ETC cards** (effective-theory cards) carrying error bars, validity regimes, and calibration records. Two gates discipline the loops: G1 (human authority over specification freezes) and G2 (every piece of physics code ships with an analytic reference test). A *weather organ* couples a barotropic toy core to learned corrections trained online (through-solver SPSA); a *flood organ* (Lax–Friedrichs shallow-water) couples rain to inundation and loss. None of this architecture is the subject of this paper — the *examination* of it is. For the full design, see the project whitepaper (§4, ARM architecture design).

3 ScaleBench: a benchmark for auto-renormalization

3.1 Design: five families, five failure modes

Each task family targets one canonical failure of learned effective theories, and each task’s anti-gaming mechanism is public:

- **Known-Recovery** — recover a known law (KPP pulled-front speed $c = 2\sqrt{rD}$), scored down to the prefactor: finite-difference front lag is charged as measured error, so “looks right” does not score.
- **Cross-Resolution** — the same law must come out of three grids with the same prefactor; grid-overfit laws fail by construction.
- **Long-Horizon** — a closure is audited for drift over $100\times$ the training horizon; uncalibrated closures necessarily score low.
- **Discovery** — no textbook answer exists: the system must emit a macroscopic equation *with error bars and regime boundaries* (including an explicit `theory_gap` field), arbitrated by held-out data.
- **Extrapolation** — evaluation outside the calibration domain; interpolation systems score perfectly inside and collapse outside. This is the law-vs-interpolation watershed.
- **Chaos-Audit** — Lyapunov exponents and attractor distances enter the acceptance gate: cards are decision-eligible only if chaos constraints are measured, not assumed.

3.2 Leaderboard (v2.3, deterministic pipeline, seed-fixed)

3.3 Anti-gaming is measured, not asserted

When the turbulence-spectrum task was first set to “recover the textbook k^{-3} ”, measurement overthrew the task design itself (all regimes measured -2.0 to -2.2); the task was corrected to regime-effective bands, and the episode is on record in the task metadata. The scoreboard is allowed to be wrong — and to show the correction.

3.4 Executable arbitration

`python -m scalebench.reproduce` reruns the full deterministic pipeline and compares each task against the public leaderboard within ± 0.02 (cross-platform float tolerance), exiting nonzero on any mismatch. Full reproduction is a *precondition* for external submissions; claimed error-bar coverage is checked by binomial z -test; the Discovery family is the final arbiter. The adjudication instrument is itself adjudicable.

4 The live AIFS comparison harness

4.1 Archive with byte-level provenance

ECMWF’s open-data AIFS feed is a rolling window (measured coverage ≈ 4 days, correcting an initial estimate of ~ 30). We therefore archive operationally: key steps (0/12/24/48/72/96/120 h) of both daily runs are captured continuously since 2026-09-17, each file registered with sha256 and byte count; the manifest verifier detects tampering, deletion, and unregistered additions. Any conclusion below can be traced to the exact bytes it was computed from.

4.2 Measured deterministic baselines (rolling)

Truth caliber, stated plainly: the verification reference is AIFS’s own later 0 h field — a *self-consistency* caliber (forecast vs. own analysis frame), not independent-observation accuracy. Run-to-run spread is recorded as first-class data (single-run readings are not verdicts).

4.3 Precipitation ETS channel: two convictions, then the first sample

The first *independent-truth* comparison (AIFS 24 h accumulated precipitation vs. ERA5 daily totals, PRD window, multi-threshold ETS) landed on 2026-09-24 once ERA5T latency cleared (run 2026-09-18 00z, verified against the 2026-09-18 daily total): AIFS window mean 0.34 mm vs. ERA5 0.38 mm; multi-threshold ETS = 0.712 (0.1 mm), 0.241 (1 mm), 1.0 (5 mm and 10 mm — a small-rain event, so the high thresholds are trivially saturated and we say so). Two further samples landed on 2026-09-25 (runs 2026-09-18 12z and 2026-09-19 00z vs. the 2026-09-19 daily total): ETS at 0.1 mm = 0.830 / 0.495 and at 1 mm = 0.176 / 0.143. A fourth landed on 2026-09-26 (run 2026-09-20 00z vs. the 2026-09-20 daily total — the first real rainband case, window means 1.87 vs. 1.61 mm): ETS = 0.691 (0.1 mm), 0.609 (1 mm), 0.221 (5 mm — the first non-trivial high-threshold reading), 1.0 (10 mm).

A third conviction: time-axis caliber. On 2026-09-26 the channel caught its own calibration error. ERA5 daily files are *accumulation-ending* (day d covers $[d-1\ 23:00, d\ 23:00)$), so a 00z+24h forecast aligns only with the *run-day* file (± 1 h); an apparently legitimate “verify-day” pairing is a ~ 23 h offset. One such record (ETS 0.238, initially logged as a genuine rainband miss) was convicted by exact-window hourly reconstruction (ETS 0.501, matching the run-day pairing 0.495) and is retained as RETRACTED evidence; both guards now reject such pairings in code, and 12z runs are routed to minute-aligned exact-window reconstruction (first sample: 0.840). Valid samples now stand at $N=4$ in two calibers (run-day ± 1 h: 0.712/0.495/0.691; exact-window: 0.840), and the verdict standard itself is drafted, not frozen (DRAFT-SPEC-008, pending human ratification).

A saturated-threshold conviction. The fourth sample (2026-09-21, the first heavy-rain case: AIFS 2.88 vs. ERA5 2.60 mm window means) scored ETS = 0 at the 0.1 mm threshold — not a miss: AIFS reported rain at *every* window grid point, so all observed events were “hit” and yet the chance reference equals the hit count exactly (zero skill beyond random). At 1 mm the same pair scores 0.642. The multi-threshold design carries its own diagnosis: drizzle-threshold saturation is visible in the table, not hidden by a headline number.

The first verdict (2026-09-28). With $N=5$ in the run-day pool {0.712, 0.691, 0.495, 0.088, 0.000}, the DRAFT-SPEC-008 trial verdict landed — **V-INCONCLUSIVE**: median 0.495, a hair (0.005) below the V-PASS threshold, with the two wet-day drizzle scores pulling the median away from the dry-day band. The first exercise of the standard produced a boundary case immediately — which is what a discriminating standard is for. Two ratification questions are now on the table (G1-8: stratify by wet/dry regime; G1-9: median robustness under semantically different zeros). Before the sample landed, the pipeline had already caught two real defects, both now guarded by tests: (i) a $1000\times$ unit contamination — AIFS GRIB **tp** is natively kg m^{-2} (=mm) while ERA5 NetCDF **tp** is in metres; the two readers now pass through an explicit unit-normalization that raises on unknown units rather than assuming; (ii) a *cross-year mispairing* in one of our own runs (a 2026 forecast scored against a 2018 typhoon day), which produced a plausible-looking but meaningless “AIFS misses the rain band” result. That record is marked RETRACTED in the public JSON (evidence retained), and the runner now hard-rejects a verification day outside the forecast window. We regard the retraction as a feature of the method, and we kept the receipt.

4.4 Quasi-stationary planetary waves: a model-truth drift match

A five-day ERA5 vo500 sequence (2026-09-16 to 20, mid-latitude band) shows all zonal modes $m=3$ –8 quasi-stationary (phase drift $< 15\%$ of the free Rossby–Haurwitz rate; the 18-h sub-window’s strong $m=6$ drift is transient — the long/short window contrast is itself reading discipline). The two-layer QG model, without any orographic term, reproduces the $m=3$ drift to within 33% ($+6.3 \times 10^{-6}$ vs $+4.8 \times 10^{-6}$ rad/s) through the jet Doppler–Rossby balance alone; an idealized continent does not anchor further (10^{-3} accelerates, 10^{-2} destabilizes). Quasi-stationarity here is a jet-balance phenomenon, not an orographic one — recorded as ScaleBench task `rossby_quasi_stationary`.

4.5 ENS status

The CRPS metric against AIFS-ENS is registered but unmeasured: all five candidate URL patterns for ensemble products on the open store return 404 (probed 2026-09-21, rate-limit backoff respected), while deterministic products return 206 for the same run. Ensemble verification requires an ECMWF account (MARS) or future open releases.

5 Real-data training rings: protocol works, deficit is dynamical

The weather organ’s real-data chain (rings #4–#7) trains a pointwise correction on tendency residuals of an ERA5-derived vorticity series, then fine-tunes through the solver (SPSA on multi-step rollout loss). Grid: 13×13 barotropic core, 6 h steps, normalized anomaly units.

5.1 The protocol improves on its own objective on both truth channels

Geostrophic vorticity from z_{500} (ζ_g) and official ERA5 vorticity vo@500 are two independent truth channels; the official field is the amplitude-truth channel (geostrophic Laplacian inflates grid noise $\sim 6\times$: measured RMS ratio 0.164). On *both* channels the online stage rescues the offline closure and beats free-run and persistence baselines:

5.2 The deficit is dynamical, not spectral (a registered negative result)

Why is skill on the cleaner vo channel *worse* (1.348) than on ζ_g (1.043)? The scale hypothesis — vo contains small-scale content the toy core cannot represent, so filtering the truth should recover skill — was tested directly (isotropic spectral low-pass, analytic reference tests: mode pass/kill, Parseval accounting, idempotence) and *rejected by measurement*: low-passed truth scored worse, not better (online 1.369 vs. 1.348 unfiltered), and the free-run skill curve degrades monotonically as the cutoff tightens. The deficit is therefore not about wavelength. The pointed candidate — that vo carries off-manifold (unbalanced, inertia–gravity) content the barotropic core cannot evolve — was itself examined by a balance-manifold projection audit (ring #7): official vo is found to lie essentially *on* the balanced manifold (projection $\approx$ identity, energy ratio > 0.99), i.e. the remaining error is dominated by the toy core’s own dynamical poverty, not by truth-manifold mismatch. Each of these steps is pinned by guard assertions so future silent rewrites of the conclusion turn the suite red. Negative results are registered with the same standing as positive ones.

6 Physicalized rain–flood coupling

Until this season, per-case peak normalization pinned every typhoon case’s rain peak to the same nondimensional value — erasing the fact that Hato’s 26.7 mm/6h and Saola’s 22.4 mm/6h are not

the same rain. We abolished it: ERA5 `tp` now drives the shallow-water organ in physical millimetres through an explicit mapping ($R_{nd} = r_p L_p / (H_p^{3/2} \sqrt{g_p})$; L_p fixed by window geography, H_p calibrated *once* across cases from the calibration ring’s anchor and shared thereafter). Guard assertions encode the anti-normalization invariant: cross-case rain peaks must order by measured mm (Hato > Pakhar, ratio 2.00 vs. measured `tp` ratio 2.01). The calibration record, including its $\pm 10\%$ per-case residuals, is shipped as data; absolute-depth claims carry that calibration uncertainty explicitly. End-to-end analytic anchoring: uniform physical rain on a flat dry bed raises depth by exactly $r_p \times$ elapsed physical time through the full nondimensional–metric round trip (10^{-6} relative).

A twelve-case comparison against the old parametric bridge quantifies what the bridge’s spatial proxy was worth: peak-flood pattern correlation -0.64 to 0.26 (one case anti-correlated), bridge domain-mean depth $2.9\text{--}15.2\times$ too wet in all twelve cases, and loss overestimation in the same direction everywhere (relative deltas explode to 10^{18} in weak-rain cases where the `tp`-driven loss is near zero — there the *absolute* error is the meaningful reading, and it is the bridge that is measured, not the truth).

7 Discipline as method

Five rules did the quiet work in this paper: (1) no measured number is deleted; failures are registered in a known-failures ledger, thresholds are never loosened to pass; (2) every physics module ships an analytic reference test (G2) — hand-computed anchors, Parseval identities, exact uniform-rain solutions; (3) trigger-line tests that turn red when assumptions change are upgraded to true assertions, never deleted; (4) partial data is skipped honestly (runtime errors with remediation), never back-filled; (5) retraction is a first-class outcome — one is on public display in the precipitation ledger. We are aware of how this reads in a field driven by leaderboard deltas; we mean it as the differentiator.

8 Artifacts and rolling updates

- Public page (leaderboard + measured baselines, rolling): <https://dolo-borigjin.github.io/The-everything/>
- Repository (MIT): <https://github.com/Dolo-borigjin/The-everything>
- Archived release with DOI (Zenodo, concept DOI resolves to latest): <https://doi.org/10.5281/zenodo.22933596>
- Single source of truth: `leaderboard.json`; arbitration: `python -m scalebench.reproduce` (nonzero exit on mismatch $> \pm 0.02$).
- Measured baselines roll onto the page via an idempotent updater (`update_measured.py`, freshness-checked in CI); archive manifests are sha256-verified per file.
- The 12-case ERA5 typhoon replay (2016–2025, `tp`-driven flood chain, physical rain colormap in mm/6h) ships as an interactive demo in the project workspace; its data pipeline is assertion-guarded end to end.

9 Limitations and honest boundaries

The toy core is barotropic and 13×13 ; nothing here is yet a global engine. Most baseline rows use the self-consistency caliber (own-analysis truth); independent-truth verification reached $N=3$ valid precipitation pairs as of 2026-09-25 (§4.3) and accumulates on the rolling archive. ENS comparison awaits ECMWF-side availability. Absolute depths in the flood organ carry calibration-regime uncertainty. The interesting question — does the adjudication-first method keep paying once the engine scales — is exactly what the next season measures.

10 Next season (rolling)

the precipitation ETS stream accumulating to weekly cadence (first sample landed 2026-09-24); ensemble CRPS once ENS is reachable; manifold-native law discovery on the vo truth channel; and the preprint you are reading, updated as the numbers move.

References

- [1] ECMWF, *AIFS: Artificial Intelligence Forecasting System*, model documentation and AIFS blog series, 2025–2026.
- [2] Lam, R. et al., *Learning skillful medium-range global weather forecasting* (GraphCast), Science, 2023.
- [3] Bi, K. et al., *Accurate medium-range global weather forecasting with 3D neural networks* (Pangu-Weather), Nature, 2023.
- [4] Kochkov, D. et al., *Neural general circulation models for weather and climate*, Nature, 2024.
- [5] Deng, J. et al., *ImageNet: A large-scale hierarchical image database*, CVPR, 2009.

Family	Task	Score	Key measured detail
Known-Recovery	known_recovery_kpp	0.998	exponent error 0; prefactor error 0.004 after Ebert–van Saarloos asymptotic extrapolation (was 0.290 via naive secant measurement)
Cross-Resolution	cross_resolution_kpp	1.000	prefactor spread 0.000 over 3 grids
Long-Horizon	longhorizon_closure_drift	1.000	drift audit passed; uncalibrated closures fail by design
Discovery	spectrum_law_turb2d	1.000	effective slope -2.03 ± 0.05 ($R^2 = 0.98$), dual cascade; gap to asymptotic k^{-3} reported
Discovery	m3_partition_law	0.976	partition law $\gamma = 0.7927 \pm 0.0008$ found <i>from the conservation audit ledger</i>
Extrapolation	extrapolation_kpp	0.960	6 out-of-domain configurations, mean relative error 1.60%
Discovery	nis_prune_consistency	1.000	data-driven dimension k ; order parameter reconstructible (ratio 0.9991)
Chaos-Audit	chaos_audit_gate	0.994	$\lambda = 0.69295$ vs. $\ln 2$ (rel. err. 2.9×10^{-4}); 3 regime verdicts correct
Discovery	rg_invariant_sir	0.997	attack-rate ε -collapse vs. analytic final size (0.34%); peak height/time correctly judged non-invariant
Long-Horizon	closure_rescue_sir	1.000	online SPSA objective -58.8% ; held-out rescue margin 23.6% over free run; error $2.64\times$ oracle ceiling (self-reported)
Extrapolation	extrapolation_turb2d	1.000	ν doubling $\times 2$: slope $-2.019 \rightarrow -2.101 \rightarrow -2.265$ monotone steepening (probe $R^2 \geq 0.98$); per-probe deviation reported
Long-Horizon	closure_switch_rescue_sir	0.971	policy switch mid-epidemic: post-switch loss 0.643 vs frozen 4.820; anti-vacuity guard self-certified; stability rel-diff 9.7×10^{-5}
Long-Horizon	regime_shift_adaptation	1.000	switch family $t_s=2/3/4$: gated-RLS mean improvement 38.1%, worst 0.0% (gate closed when calm); ungated RLS wrecks $t_s=2$ by $-1.9\times$
Discovery	qg2_modal_growth	0.987	baroclinic jet eigenvalue $1.12 \times 10^{-5}/s$ @ $m=3$; Benettin/eigen ratio 0.96; barotropic control stable; real-basis convention convicted
Long-Horizon	decision_tiers_sir	1.000	preemptive 0.415 < reactive 0.627 < deployed feedback 0.903; forecast value = 0.212 cost units; MPC responds to $4\times$ weight switch
Chaos-Audit	ets_caliber_guard	1.000	three rejection classes (23h/11h/cross-year) + exact-window arithmetic + saturation self-diagnosis
Chaos-Audit	ensemble_bred_qg2	0.845	bred direction at $0.85\times$ modal rate; spread grows 0.90–0.93 $\times$ daily; $s/s=9.4$ over-dispersive reported (half credit)
Discovery	rossby_quasi_stationary	1.000	jet Doppler-RH drift $+6.3 \times 10^{-6}$ vs ERA5 five-day quasi-stationary $+4.8 \times 10^{-6}$ rad/s (33%); orographic non-anchoring on record
Overall		0.9864	

Table 1: ScaleBench v2.3 leaderboard (17 tasks; two further growth-protocol and ENIAC-recreation rows omitted from this table for space; leaderboard.json is canonical) (single source of truth: leaderboard.json). Sub-unity scores are recorded, not hidden: 0.826 was a measurement artifact, not physics: the FD pulled-front lag turned out to be an algebraic transient of pulled fronts (not discretization), and was corrected by an Ebert–van Saarloos asymptotic estimator whose log-coefficient carries the physical signature $b = -1.5\sqrt{D/r}$ — prefactor error $0.290 \rightarrow 0.004$, score $0.826 \rightarrow 0.998$ (2026-09-25, episode on record in the test suite). What remains sub-unity is recorded as physical difficulty — the floor of the score is the difficulty of the physics.

Lead	z_{500} RMSE global (m^2/s^2)	z_{500} RMSE PRD (m^2/s^2)	MSL RMSE global (Pa)	MSL RMSE PRD (Pa)
+12 h	31.9 [30.2–34.5, $n=4$]	21.1 [4.8–30.4]	73.5 [64.9–80.7]	25.0 [13.5–38.8]
+24 h	46.6 [40.5–50.0, $n=5$]	22.9 [12.4–33.7]	102.9 [95.4–109.3, $n=6$]	26.2 [21.7–31.4]
+48 h	75.3 [69.4–87.7, $n=4$]	18.1 [11.7–24.7]	139.5 [127.2–156.7]	31.9 [25.5–42.1]
+60 h	89.9	18.2	145.0	24.7
+72 h	120.1 [105.6–149.8, $n=4$]	23.2 [17.5–27.9]	179.9 [162.0–203.4]	24.4 [17.3–29.6]
+84 h	129.5	18.4	181.9	26.6
+96 h	176.4 [170.8–181.0, $n=3$]	25.9 [19.7–33.5]	234.3 [222.7–243.2, $n=3$]	44.2 [25.6–59.0]
+120 h	259.2 [253.7–267.1, $n=3$]	41.3 [39.2–43.6]	308.0 [290.0–317.8, $n=3$]	27.4 [16.9–40.7]

Table 2: Measured AIFS-single deterministic baselines (as of 2026-09-25; mean over runs with [min–max, n] where $n > 1$). Degradation with lead is monotone in the global mean through 120 h ($49 < 72 < 90 < 129 < 150 < 176 < 214 < 259 \text{ m}^2/\text{s}^2$, multi-run means); window-level wiggles are physical and recorded. The table rolls forward on the public page (§8).

Truth channel	Persistence	Free run	Offline closure	Online closure
ζ_g (ring #4)	1.264	1.219	1.313	1.043
vo@500 (ring #5)	1.979	1.948	1.574	1.348
vo@500, low-pass $k \leq 4$ (ring #6)	2.023	2.030	1.603	1.369

Table 3: 6 h anomaly skill (RMSE/std, lower is better) on the Mangkhut-2018 window. SPSA improves its own objective on both channels (loss -31% and -20%). Absolute skills > 1 are recorded, not thresholded away: the landfall window is violent, and this ring examines whether the *training protocol* works, not weather-forecast skill.